

Solution to Question 6

- (a) **Confidence Interval for p** Given: $n = 300, x = 210, \hat{p} = \frac{210}{300} = 0.70$.

Confidence Interval for a Population Proportion

$$\begin{aligned} & \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \\ & 0.70 \pm 1.96 \sqrt{\frac{0.7 \times 0.3}{300}} \\ & 0.70 \pm 0.0519 \\ & \text{Interval: } (0.6481, 0.7519) \end{aligned}$$

- (b) **Interpretation** We are 95% confident that the true proportion of all UCF students who support extending library hours until 2:00 AM is between 64.81% **and** 75.19%.